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VEHICLE CONTROL METHOD AND VEHICLE CONTROL DEVICE

Abstract

In a vehicle control device **10,** in an autonomous driving mode, when an obstacle is detected in front of a vehicle **1** in a moving direction based on an external situation acquired by a sensor **14** while the vehicle **1** is moving from a predetermined position to a parking target position by a first autonomous driving along a teacher route R**1,** it is switched to a temporary manual driving mode different from a manual driving mode, and an operation device **20** receives a manual operation by an occupant. Then, in the vehicle control device **10,** after the vehicle **1** is moved by the manual operation, when the obstacle is not detected in front of the vehicle **1** in the moving direction based on the external situation acquired by the sensor **14,** the display device **22** outputs a first screen **30**C representing return to the teacher route by the autonomous driving. Then, the vehicle control device **10** performs second autonomous driving to a halfway point between the predetermined position and the parking target position on the teacher route, and thereafter performs third autonomous driving along the teacher route from the halfway point to the parking target position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-036615, filed on Mar. 11, 2024 and Japanese Patent Application No. 2025-020878, filed on Feb. 12, 2025; the entire contents of which are incorporated herein by reference.
BACKGROUND OF THE INVENTION
1. Field of the Invention
[0002] The present disclosure relates to a vehicle control method and a vehicle control device.
2. Description of the Related Art
[0003] In general, many houses have narrow parking spaces, making parking difficult in some cases. Therefore, there is a high demand for autonomous driving to perform parking and vehicle retrieval operations.
[0004] A vehicle control device that realizes this type of autonomous driving is known. For example, a technique has been disclosed in which a vehicle is driven from a predetermined position to a parking target position by a manual operation of a driver, a movement route at that time is stored in advance as a teacher route, and after that, the vehicle is autonomously driven along the teacher route in a parking scene (refer to Patent Literature 1).
[0005] However, during autonomous driving along the teacher route, there may be an obstacle on the teacher route that was not present during the teacher driving. In the related art, in such a case, autonomous parking by autonomous driving may be difficult. That is, in the related art, it may be difficult to provide suitable parking assistance.
[0006] The present disclosure has an object to provide a vehicle control method and a vehicle control device capable of providing more suitable parking assistance.
SUMMARY OF THE INVENTION
[0007] A vehicle control method according to the present disclosure is executed by a vehicle that includes a sensor that acquires an external situation, an operation device that receives an operation by an occupant, a steering device that receives a steering operation by the occupant, a display device that is visually recognizable by the occupant, and a movement controller that controls at least steering, the vehicle holding a teacher route obtained by teacher driving from a predetermined position to a parking target position. In the vehicle control method according to the present disclosure, in a manual driving mode, the vehicle is driven in response to a steering operation on the steering device by the occupant, and when the vehicle is within a predetermined range from the predetermined position, and the operation device receives a first operation, switching to an autonomous driving mode is performed, thereafter, in the autonomous driving mode, when an

obstacle is detected in front of the vehicle in a moving direction based on an external situation acquired by the sensor during movement from the predetermined position to the parking target position along the teacher route by first autonomous driving controlling at least the steering, switching to a temporary manual driving mode different from the manual driving mode is performed, and the operation device and the steering device receive a manual operation by the occupant, thereafter, in the temporary manual driving mode, after the vehicle is moved by the manual operation, when the obstacle is not detected in front of the vehicle in the moving direction based on the external situation acquired by the sensor, the display device outputs a first screen representing return to the teacher route by autonomous driving, after second autonomous driving is performed to a halfway point between the predetermined position and the parking target position on the teacher route, third autonomous driving is performed along the teacher route from the halfway point to the parking target position, in the manual driving mode, the vehicle is driven in response to a steering operation on the steering device by the occupant, and when the vehicle is within a predetermined range from the predetermined position and the operation device does not receive a first operation, ongoingly in the manual driving mode, the vehicle is driven to the parking target position in response to a steering operation on the steering device by the occupant, and in the temporary manual driving mode, when the operation device receives a second operation, switching to the manual driving mode is performed, and in the manual driving mode, after the vehicle is moved by the manual operation, when the obstacle is not detected in front of the vehicle in the moving direction based on an external situation acquired by the sensor, the display device does not output the first screen indicating return to the teacher route by autonomous driving.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] FIG. **1** is a block diagram illustrating an example of an overall configuration of a vehicle.
[0009] FIG. **2** is a schematic diagram illustrating an example of arrangement of an in-vehicle camera.
[0010] FIG. **3** is a schematic diagram illustrating an example of a configuration of an external appearance of the vehicle.
[0011] FIG. **4** is a diagram illustrating an example of a configuration in the vicinity of a driver's seat of the vehicle of the present embodiment.
[0012] FIG. **5** is an explanatory diagram of an example of a teacher route.
[0013] FIG. **6** is an explanatory diagram of an example of estimation processing of a current position of a vehicle executed by a controller during an autonomous driving mode.
[0014] FIG. **7** is an explanatory diagram of an example of a case where an obstacle is detected on the teacher route.
[0015] FIG. **8**A is a schematic diagram of an example of a third screen.
[0016] FIG. **8**B is a schematic diagram of an example of the third screen.
[0017] FIG. **9** is a schematic diagram of an example of a fourth screen.
[0018] FIG. **10**A is a schematic diagram of an example of a first screen.
[0019] FIG. **10**B is a schematic diagram of an example of the first screen.
[0020] FIG. **11** is an explanatory diagram of an example of generation of a return route.
[0021] FIG. **12**A is a schematic diagram of an example of a second screen.
[0022] FIG. **12**B is a schematic diagram of an example of the second screen.
[0023] FIG. **13** is a schematic diagram of an example of a fifth screen.
[0024] FIG. **14** is a flowchart illustrating an example of a flow of information processing executed by a controller in a teacher driving mode.
[0025] FIG. **15** is a flowchart illustrating an example of a flow of information processing executed

by the controller in the autonomous driving mode.
[0026] FIG. **16** is a flowchart illustrating an example of a flow of interrupt processing executed by the controller.
[0027] FIG. **17** is a block diagram illustrating a hardware configuration example of a vehicle control device.

DESCRIPTION OF EMBODIMENTS

[0028] Hereinafter, embodiments of a vehicle control method and a vehicle control device according to the present disclosure will be described with reference to the drawings.
[0029] FIG. **1** is a block diagram illustrating an example of an overall configuration of a vehicle **1**.
[0030] The vehicle **1** includes a vehicle control device **10**, a movement control device **12**, a sensor **14**, a storage device **18**, an operation device **20**, a display device **22**, and a steering device **27**.
[0031] The movement control device **12**, the sensor **14**, the storage device **18**, the operation device **20**, the display device **22**, and the steering device **27** are connected to the vehicle control device **10** to be able to exchange data or signals. That is, the vehicle control device **10** is set to be connected to be capable of communicating with at least the sensor **14**, the operation device **20**, the display device **22**, the movement control device **12**, and the steering device **27**.
[0032] The movement control device **12** controls at least steering of the vehicle **1**. The movement control device **12** is means for realizing driving, braking, and turning motions necessary for driving of the vehicle **1**. For example, the movement control device **12** includes a drive motor, a power transmission mechanism, a brake device, a steering device, and the like, and an electronic vehicle control device that controls the drive motor, the power transmission mechanism, the brake device, the steering device, and the like. The movement control device **12** generates power with the drive motor and transmits the power to wheels via the power transmission mechanism to drive the vehicle **1**. The power transmission mechanism is, for example, a propeller shaft, a differential gear, a drive shaft, or the like.
[0033] Controlling at least steering means that the movement control device **12** controls at least one of driving, braking, and turning motions necessary for driving of the vehicle **1**. That is, controlling steering means that the movement control device **12** controls at least one of a turning direction by steering, a vehicle speed and acceleration by accelerator steering, and deceleration and stop by brake steering. Controlling at least acceleration and deceleration means that the movement control device **12** controls at least one of acceleration and deceleration of the vehicle **1**.
[0034] Specifically, the movement control device **12** includes an auxiliary control device **12**A, a brake control device **12**B, an engine control device **12**C, and a power steering control device **12**D. The brake control device **12**B, the engine control device **12**C, and the power steering control device **12**D can be collectively referred to as an actuator controller that controls the operation of the vehicle **1**.
[0035] The auxiliary control device **12**A is a control device that monitors the transmission state of the vehicle control device **10** and operates to execute appropriate degeneration control as a backup when the vehicle control device **10** fails. Note that even when the vehicle control device **10** fails, when safety can be secured by providing a degeneration control function in the vehicle control device **10**, the degeneration control is unnecessary.
[0036] The brake control device **12**B is a control device that performs brake control of the vehicle **1**. The brake control may be referred to as braking force control. For example, the brake control device **12**B performs brake control of the vehicle **1** in accordance with enhancement and relaxation of the operation of the brake pedal by the occupant. Specifically, the enhancement of the operation of the brake pedal by the occupant means that the occupant steps on the brake pedal. Furthermore, the brake control device **12**B performs brake control according to the surrounding video during autonomous driving.
[0037] The engine control device **12**C is a control device that controls an engine that generates a driving force of the vehicle **1**. The power steering control device **12**D is a control device that

controls power steering of the vehicle **1**.
[0038] The sensor **14** is mounted on the vehicle **1** and acquires at least the external situation of the vehicle **1**. Specifically, the sensor **14** is various sensors that detect a driving state of the vehicle **1** and the external situation of the vehicle **1**. The external situation includes video of the outside of the vehicle **1**.
[0039] The sensor **14** includes at least a camera **16**. Furthermore, the sensor **14** includes at least one of light detection and ranging (LiDAR), radar, sonar, an ultrasonic sensor, and the like. The sensor **14** includes, for example, an accelerator opening sensor that detects an accelerator opening, a steering angle sensor that detects a steering angle of a steering device, an acceleration sensor that detects acceleration acting in the front-rear direction of the vehicle **1**, a torque sensor that detects torque acting on a power transmission mechanism between wheels of the vehicle **1** and a drive motor, a vehicle speed sensor that detects a vehicle speed of the vehicle **1**, a wheel speed sensor, a global positioning system (GPS), and the like. The sensor **14** outputs sensor information obtained by the detection to the vehicle control device **10**.
[0040] The camera **16** is a surrounding sensor that is mounted on the vehicle **1** and monitors the surrounding environment of the vehicle **1**. In the present embodiment, the camera **16** captures video of the surroundings of the vehicle **1** and outputs captured video data to the vehicle control device **10**. Hereinafter, the captured video data may be simply referred to as a video. Furthermore, in the present embodiment, the camera **16** is also applied to an application of detecting an object present around the vehicle **1** and estimating the position where the vehicle **1** is present from the positional relationship between the vehicle **1** and the object present around the vehicle **1**.
[0041] The position, the number of installations, and the photographing direction of the camera **16** are adjusted in advance such that the periphery of the vehicle **1** can be photographed. For example, the vehicle **1** is provided with four cameras **16** arranged to be able to capture images in four directions of a front direction, a rear direction, a left direction, and a right direction of the vehicle **1**. Note that the number of cameras **16** provided in the vehicle **1** is not limited to four.
[0042] The storage device **18** stores various data. In the present embodiment, the storage device **18** stores data such as teacher route data **18**A and map data **18**B. That is, the storage device **18** holds the teacher route represented by the teacher route data **18**A. Details of the teacher route data **18**A and the map data **18**B will be described later.
[0043] The storage device **18** is, for example, an auxiliary storage device such as a hard disk drive (HDD), a solid state drive (SSD), or a flash memory. Note that at least a part of the data included in the storage device **18** may be stored in an external storage device such as a server device provided outside the vehicle **1** and connected to be capable of communicating with the vehicle control device **10**.
[0044] The operation device **20** receives an operation by an occupant of the vehicle **1**. The operation device **20** includes an operation mechanism related to a driving operation such as an accelerator pedal, a brake pedal, a blinker lever, and a push-in switch, and an input device such as a keyboard, a touch panel, and a switch.
[0045] At least one of the keyboard, the touch panel, and the switch functions as an autonomous parking instruction unit that receives an autonomous parking start instruction.
[0046] A shift lever is an example of a forward-backward movement operation unit. The forward-backward movement operation unit switches at least between forward movement and backward movement of the vehicle **1**. The brake pedal is an example of a brake operation unit. The brake operation unit is an operation unit for a brake that suppresses the speed of the vehicle **1**. That is, the brake operation unit receives an instruction to decelerate the vehicle **1**. The operation device **20** may constitute a part of a human machine interface (HMI) or an in-vehicle infotainment (IVI).
[0047] The display device **22** is a display that outputs various images. The display device **22** is disposed at a position visually recognizable by the occupant. The display device **22** is installed at a position visually recognizable by the occupant of the vehicle **1**. Examples of the display include a

liquid crystal display (LCD), an organic electro-luminescence (EL) display, and a projector. The display may be a touch panel display in which the display device **22** and the operation device **20** are integrally configured. The display device **22** is an example of at least one of the HMI and the IVI.
[0048] Note that the display device **22** is not limited to one including only one display area. For example, the display device **22** may include a plurality of display areas.
[0049] In addition, the vehicle **1** may include a plurality of display devices **22**. For example, the display device **22** may include a first display unit and a second display unit, and the first display unit and the second display unit are displays that output various images. The first display unit and the second display unit are display devices **22** configured as separate bodies. The first display unit and the second display unit may be disposed at different positions in the vehicle **1**. For example, the first display unit may function as the IVI, and the second display unit may function as a part of the instrument panel of the vehicle **1**.
[0050] The steering device **27** receives steering by the occupant of the vehicle **1**. The steering device **27** is, for example, a steering wheel. The steering wheel may be referred to as a handle. The steering angle of the steering device is adjusted by the operation of the steering device by the occupant.
[0051] FIG. **2** is an explanatory diagram illustrating an example of arrangement of the sensor **14** and the camera **16**.
[0052] The vehicle **1** is provided with, for example, four cameras **16** (cameras **16**A to **16**D) to be able to acquire an external situation of the vehicle **1**, for example, in at least four directions: the front, rear, right, and left directions.
[0053] Specifically, for example, the camera **16** includes a camera **16**A, a camera **16**B, a camera **16**C, and a camera **16**D. The camera **16**A is disposed in a front portion of the vehicle **1** and captures an image of the front of the vehicle **1**. The camera **16**A may be referred to as a front camera. The camera **16**B is disposed on the right side of the vehicle **1** and captures an image of the right side of the vehicle **1**. The camera **16**C is disposed on the left side of the vehicle **1** and captures an image of the left side of the vehicle **1**. The camera **16**D is disposed in a rear portion of the vehicle **1** and captures an image of the rear of the vehicle **1**. The camera **16**D may be referred to as a rear camera, a rear portion camera, or the like.
[0054] Note that the number of cameras **16** provided in the vehicle **1** is not limited to four. In addition, regarding sensors that detect objects such as a lidar, a radar, a sonar, and an ultrasonic sensor included in the sensor **14**, it is preferable that the arrangement positions, the number of arrangements, and the like are adjusted in advance such that the external situations of the right, left, front, and rear sides of the vehicle **1** can be acquired. For example, as illustrated in FIG. **2**, the sensor **14** includes sensors **14**A to **14**F. These sensors **14**A to **14**F are disposed in the vehicle **1** to be able to acquire external situations of the right, left, front, and rear sides of the vehicle **1**. Note that the sensor **14** that detects an object, such as a lidar, a radar, a sonar, or an ultrasonic sensor, may be disposed only in the rear portion of the vehicle **1**.
[0055] Next, a configuration of the vehicle **1** will be described.
[0056] FIG. **3** is a schematic diagram illustrating an example of a configuration of an external appearance of the vehicle **1**.
[0057] The vehicle **1** includes a vehicle body **2** and two pairs of wheels **23** arranged along a predetermined direction on the vehicle body **2**. The two pairs of wheels **23** include a pair of front tires **23**F and a pair of rear tires **23**R (refer to also FIG. **2**). FIGS. **2** and **3** illustrate an example in which the vehicle **1** includes four wheels **23**. However, the number of wheels **23** provided in the vehicle **1** is not limited thereto.
[0058] Next, a configuration in the vicinity of the driver's seat of the vehicle **1** of the present embodiment will be described.
[0059] FIG. **4** is a diagram illustrating an example of a configuration in the vicinity of a driver's

seat **24**A of the vehicle **1** of the present embodiment.
[0060] The vehicle **1** includes a driver's seat **24**A and a passenger seat **24**B. A windshield **25**, a dashboard **26**, the steering device **27**, an operation button **20**B, and the display device **22** are provided in front of the driver's seat **24**A. A shift lever **20**C, which is a lever for shifting the transmission, is provided in the vicinity of the driver's seat **24**A.
[0061] The steering device **27**, the operation button **20**B, and the shift lever **20**C are examples of the operation device **20**.
[0062] The steering device **27** is provided in front of the driver's seat **24**A and can be operated by an occupant. The steering device **27** receives the steering operation of the occupant as described above. The rotation angle of the steering device **27**, that is, the steering angle is electrically or mechanically interlocked with the change in the direction of the front tires **23**F, which are steering wheels. The steering wheels may be the rear tires **23**R, or both the front tires **23**F and the rear tires **23**R may be the steering wheels.
[0063] The operation button **20**B is a button capable of receiving an operation by the occupant. The operation button **20**B may include a direction indicator. The position of the operation button **20**B is not limited to the example illustrated in FIG. **4**, and may be provided in the steering device **27**, for example. Although one operation button **20**B is illustrated in FIG. **4**, a plurality of operation buttons **20**B may be provided. When the display device **22** also serves as a touch panel, the display device **22** may be an example of the operation device **20**.
[0064] Returning to FIG. **1**, the description will be continued.
[0065] The vehicle control device **10** is an electronic control unit that integrally controls each unit of the vehicle **1**.
[0066] The vehicle control device **10** controls the movement control device **12** such that the driving state of the vehicle **1** is optimized using the sensor information, the captured video, and the like received from the sensor **14** and the camera **16**, respectively. In addition, the vehicle control device **10** controls the movement control device **12** to autonomously drive the vehicle **1**.
[0067] The vehicle control device **10** includes a controller **11**. A part or all of the controller **11** may be a software configuration realized by cooperation of a processor and various programs stored in a memory. In addition, a part or all of the controller **11** may have a hardware configuration realized by a dedicated circuit or the like.
[0068] The controller **11** integrally controls each unit of the vehicle **1**.
[0069] In the present embodiment, the controller **11** is configured to be able to switch the driving mode to a teacher driving mode, an autonomous driving mode, a manual driving mode, or a temporary manual driving mode based on an input operation or the like of the operation device **20** by the occupant. The driving mode executable by the vehicle **1** may include various driving modes other than the teacher driving mode, the autonomous driving mode, the manual driving mode, and the temporary manual driving mode.
[0070] The teacher driving mode is a mode for registering a teacher route when the vehicle **1** is driven autonomously. The teacher route is a route obtained by teacher driving from a predetermined position to a parking target position. In the teacher driving mode, the vehicle **1** is controlled to be driven by the manual operation of the occupant. That is, in the teacher driving mode, the controller **11** controls the movement control device **12** to drive the vehicle **1** in response to the manual operation that is a driving operation by the occupant.
[0071] The autonomous driving mode is a mode in which the vehicle **1** is driven autonomously. In the present embodiment, the autonomous driving mode means a mode in which the vehicle **1** is driven autonomously along the teacher route. In the autonomous driving mode, the controller **11** controls at least steering to control the movement control device **12** to drive the vehicle **1** along the teacher route. In the autonomous driving mode, the vehicle **1** is autonomously controlled to be driven by the vehicle control device **10** without the manual operation by the occupant.
[0072] The manual driving mode is a driving mode in which the vehicle **1** is driven in response to

the steering operation on the steering device **27** by the occupant.
[0073] The temporary manual driving mode is a driving mode in which manual driving is temporarily executed in the autonomous driving mode. The temporary manual driving mode is a driving mode different from the manual driving mode. Specifically, the temporary manual mode is a driving mode in which the operation device **20** and the steering device **27** temporarily receive the manual operation by the occupant to drive the vehicle **1** during autonomous driving, and then the vehicle **1** is driven autonomously when a predetermined condition is satisfied.
[0074] FIG. **5** is an explanatory diagram of an example of a teacher route R**1**.
[0075] In the teacher driving mode, the teacher driving from a predetermined position P**1** to a parking target position P**2** is performed by the manual operation by the occupant. The parking target position P**2** is, for example, a parking lot or the like, but is not limited thereto. Further, the predetermined position P**1** may be any position of the occupant in the real space.
[0076] The driving route R through which the teacher driving is performed is treated as the teacher route R**1**, and the teacher route data **18**A of the teacher route R**1** is stored in the storage device **18**. During the teacher driving, the occupant may perform the manual operation to drive the vehicle from the parking target position P**2** toward the predetermined position P**1**, or may perform the manual operation to drive the vehicle from the predetermined position P**1** toward the parking target position P**2**. When the vehicle **1** is driven from the parking target position P**2** toward the predetermined position P**1** in the teacher driving mode, the controller **11** may create the teacher route data **18**A of the teacher route R**1** in which the driving direction of the driving route R during the teacher driving is set to the reverse direction. When the vehicle **1** is driven from the predetermined position P**1** toward the parking target position P**2** during the teacher driving, the controller **11** may create the teacher route data **18**A of the teacher route R**1** along the driving direction of the driving route R during the teacher driving. Details of the creation of the teacher route data **18**A will be described later.
[0077] In the autonomous driving mode, the controller **11** controls at least steering along the driving route R obtained by the teacher driving to autonomously drive the vehicle **1** to the parking target position P**2**. In the autonomous driving mode, the controller **11** executes steering control of the vehicle **1** and front and rear acceleration/deceleration control, but at least a part of front and rear acceleration/deceleration control may be executed by the operation of the driver.
[0078] Next, control by the controller **11** in each of the teacher driving mode and the autonomous driving mode will be described in detail.
In Teacher Driving Mode
[0079] First, the control of the controller **11** in the teacher driving mode will be described in detail.
[0080] The controller **11** switches the driving mode to the teacher driving mode when receiving a signal indicating an instruction to start the teacher driving mode by the operation or the like of the operation device **20** by the occupant. Then, the controller **11** executes the following processing in the teacher driving mode.
[0081] The controller **11** acquires sensor information indicating a driving state of the vehicle **1** from the sensor **14**. Then, the controller **11** estimates the current position of the vehicle **1** based on a temporal change in the sensor value represented by the sensor information. For example, the controller **11** calculates a movement amount of the vehicle **1** from a reference position such as a driving start position when the teacher driving mode is started based on a temporal change in the vehicle speed and the yaw rate represented by the sensor values, and estimates the current position of the vehicle **1** based on the movement amount.
[0082] Note that the estimation accuracy of the current position based on the movement amount may be low. Therefore, the controller **11** may use, as the current position, a result of correcting the estimated current position based on the captured video around the vehicle **1** acquired by the camera **16**.
[0083] The controller **11** sequentially stores the current position of the vehicle **1** sequentially

estimated along the driving of the vehicle **1** in the storage device **18**. Specifically, the controller **11** sets, as the teacher route R**1**, the driving route R during the teacher driving represented by a group of current positions sequentially estimated from a time point when the instruction to start the teacher driving mode is received until the instruction to end the teacher mode is received, and stores the teacher route data **18**A representing the teacher route R**1** in the storage device **18**.

[0084] The teacher route data **18**A includes a group of pieces of driving information for each position which is a current position sequentially estimated during the teacher driving. The driving information includes an INDEX, a driving position, an azimuth, a driving direction, and reference driving information. The INDEX is identification information of driving information. The driving position is the estimated position of the vehicle **1**. The azimuth indicates the direction of the vehicle **1** at the position. The driving direction indicates a driving direction of the vehicle **1** at the position, and is represented by, for example, forward movement or backward movement. The reference driving information is information representing a driving state or the like at the position. The reference driving information is, for example, information such as a steering angle and a vehicle speed detected at each position during the teacher driving.

[0085] In addition, during the teacher driving of the vehicle **1**, the controller **11** creates the map data **18**B for estimating the current position of the vehicle **1** from the captured video captured by the camera **16**. As a method for estimating the current position of the vehicle **1** from the captured video, a simultaneous localization and mapping (SLAM) method or the like is used.

[0086] The map data **18**B is map data in which a plurality of feature points around the vehicle **1** at the time of driving along the teacher route R**1** are registered.

[0087] The feature point is a feature point obtained by performing image analysis on the captured video captured by the camera **16** during the teacher driving. For example, the feature point is a part where a characteristic image pattern is obtained by analyzing the captured video in an object (for example, a tree, a wall, a column, or the like) or the like that can be a mark in the real view. The part is, for example, an edge part of the object. The map data **18**B includes a plurality of feature points, and each feature point is identifiably registered for each feature point by being assigned an identification number.

[0088] The feature point is represented by feature point data including a three-dimensional position and a feature amount. The three-dimensional position of the feature point is a three-dimensional position of the feature point in the real space, and is represented by, for example, a three-dimensional orthogonal coordinate system (X, Y, Z). The feature amount of the feature point is a feature amount represented by image analysis of the captured video of the feature point. The feature amount of the feature point is, for example, luminance and density on the captured video, a scale invariant feature transform (SIFT) feature amount, a speeded up robust features (SURF) feature amount, or the like.

[0089] In the map data **18**B, one feature point is registered for each identical three-dimensional position. Note that, for the same three-dimensional position, a plurality of feature points may be registered in the map data **18**B for each photographing position and photographing direction by the camera **16** at the three-dimensional position. In addition, the feature point data of the feature points registered in the map data **18**B may further include image data of an object having the feature point.

[0090] During the teacher driving, the controller **11** specifies coordinates of feature points in the real view based on, for example, stereo photogrammetry. Specifically, the controller **11** reads a plurality of captured videos captured at different timings, and associates the same feature point commonly appearing in the plurality of captured videos. Then, for example, the controller **11** estimates a temporary position of the vehicle **1** when the plurality of captured videos are captured, and specifies temporary coordinates of the feature point in the real view by the principle of triangulation. Then, for example, the controller **11** performs bundle adjustment using the temporary position of the vehicle **1** and the temporary coordinates of the feature point in the real view as

reference information, and calculates the formal position of the vehicle **1** and the formal coordinates of the feature point in the real view to minimize a reprojection error when each feature point in the real view is projected on all the captured videos. Then, the controller **11** stores, in the storage device **18**, the map data **18**B in which the feature point represented by the feature point data including the formal coordinates of the feature point in the real view as the three-dimensional position is registered.
[0091] The three-dimensional position of the feature point registered in the map data **18**B may be a position measured in advance using LiDAR or a stereo camera without using the SLAM method. However, from the viewpoint of suppressing a decrease in position estimation accuracy, the SLAM method may be used.
[0092] As described above, the controller **11** executes the above processing in the teacher driving mode. Therefore, during the teacher driving mode, the controller **11** generates the teacher route data **18**A of the teacher route R**1** obtained by the teacher driving from the predetermined position P**1** to the parking target position P**2** and the map data **18**B in which the three-dimensional position of each of the plurality of feature points around the vehicle **1** during the driving along the teacher route R**1** and the feature amount of the feature point are registered, and stores the generated data in the storage device **18**.
Autonomous Driving Mode
[0093] Next, the control of the controller **11** during the autonomous driving mode will be described in detail.
[0094] The controller **11** switches the driving mode to the autonomous driving mode when receiving a signal indicating an instruction to start the autonomous driving mode as the operation device **20** receives the first operation by the operation or the like of the operation device **20** by the occupant. Then, the controller **11** executes the following processing in the autonomous driving mode.
[0095] The controller **11** reads the teacher route data **18**A and the map data **18**B from the storage device **18**, and controls the movement control device **12** to perform autonomous driving along the teacher route R**1** represented by the teacher route data **18**A.
[0096] The controller **11** estimates the current position of the vehicle **1** based on the map data **18**B and the captured video around the vehicle **1** acquired by at least one camera **16**.
[0097] FIG. **6** is an explanatory diagram of an example of estimation processing of the current position of the vehicle **1** executed by the controller **11** during the autonomous driving mode.
[0098] S**1**, S**2**, and S**3** in FIG. **6** represent three feature points extracted from the captured video of the camera **16**, and points Q**1**, Q**2**, and Q**3** are feature points stored in the map data **18**B, and represent three-dimensional positions of the feature points SA, SB, and SC in real space. RP**1** represents an imaging plane of the camera **16**. The point P′ represents the position (that is, the position of the vehicle **1**) of the camera **16** obtained from the three feature points SA, SB, and SC extracted from the captured video of the camera **16** and the feature points (Q**1**, Q**2**, Q**3**) stored in the map data **18**B.
[0099] For example, the controller **11** first collates the feature points extracted from the captured video of the camera **16** with the feature points stored in the map data **18**B using pattern matching, feature amount search, or the like. Then, the controller **11** randomly selects several (for example, 3 to 6) feature points among the feature points extracted from the captured video of the camera **16** and the feature points that can be collated with the feature points stored in the map data **18**B.
[0100] Then, the controller **11** estimates the current position of the vehicle **1** in the real space based on the positions of these several feature points in the captured video and the three-dimensional positions of the feature points registered in the map data **18**B corresponding to the several feature points in the real space. At this time, the controller **11** estimates the current position of the vehicle **1** by solving the PnP problem using, for example, a known method (for example, the document: Mikael Persson et al. “Lambda Twist: An Accurate Fast Robust Perspective Three Point (P3P)

Solver.", ECCV 2018, pp 334-349, published in 2018, http://openaccess.thecvf.com/content_ECCV_2018/papers/Mikael_Persson_Lambda_Twist_An_ECCV_2018_paper.pdf) such as Lambda Twist.

[0101] When collating the feature points extracted from the captured video of the camera **16** with the feature points stored in the map data **18**B, for example, the controller **11** may calculate the current position of the vehicle **1** as a temporary position based on the movement amount of the vehicle **1** described above, and may narrow down the feature points to be collated with the feature points extracted from the captured video of the camera **16** among the feature points stored in the map data **18**B with the temporary position as a reference.

[0102] Through these processing, the controller **11** estimates, as the current position information indicating the current position of the vehicle **1**, current position information including information related to the two-dimensional position (X coordinate, Y coordinate) of the vehicle **1** in the real space and the posture that is the direction of the vehicle **1** based on the map data **18**B and the captured video around the vehicle **1** acquired by at least one camera **16**.

[0103] Then, the controller **11** autonomously drives the vehicle **1** from the predetermined position P**1** toward the parking target position P**2** along the teacher route R**1** by controlling the movement control device **12** such that the estimated current position of the vehicle **1** is a position on the teacher route R**1** represented by the teacher route data **18**A. Then, the controller **11** stops the vehicle **1** at the parking target position P**2**.

[0104] When the vehicle **1** is driven autonomously along the teacher route R**1**, the controller **11** feedback-controls the movement control device **12** such that the vehicle **1** moves along the teacher route R**1** based on the estimated current position of the vehicle **1** and each position on the teacher route R**1** represented by the teacher route data **18**A.

[0105] Hereinafter, autonomous driving from the predetermined position P**1** along the teacher route R**1** to the parking target position P**2** by the vehicle **1** will be referred to as first autonomous driving.

[0106] Here, there is a case where an obstacle is detected on the teacher route R**1** while the vehicle **1** is moving along the teacher route R**1** from the predetermined position P**1** to the parking target position P**2** by the first autonomous driving.

[0107] FIG. **7** is an explanatory diagram of an example of the case where an obstacle B is detected on the teacher route R**1**.

[0108] As illustrated in FIG. **7**, there is a case where the obstacle B that is not present at the time of registration of the teacher route data **18**A is present on the teacher route R**1** during the first autonomous driving from the predetermined position P**1** to the parking target position P**2**.

[0109] The obstacle B is an object that obstructs the driving of the vehicle **1**. Specifically, the obstacle B is an object that causes difficulty in continuing driving when the vehicle **1** comes into contact with the object, or has a possibility of causing some damage to the vehicle **1** or an event that hinders driving when the vehicle **1** comes into contact with or tries to pass over the object without avoiding the object.

[0110] Therefore, in the present embodiment, the controller **11** determines whether or not the obstacle B has been detected in front of the vehicle **1** in a moving direction D based on the external situation acquired by the sensor **14** during the movement by the first autonomous driving.

[0111] Specifically, in the manual driving mode, the controller **11** drives the vehicle **1** in response to a steering operation on the steering device **27** by the occupant. Then, when the vehicle **1** is within a predetermined range from the predetermined position P**1** and the operation device **20** receives the first operation, it switches to the autonomous driving mode. Thereafter, in the autonomous driving mode, at least steering is controlled along the teacher route R**1** to perform first autonomous driving toward the parking target position P**2**.

[0112] The first operation may be a predetermined operation indicating an instruction to start the autonomous driving mode.

[0113] The predetermined range from the predetermined position P**1** may be within a

predetermined range from the predetermined position P**1**. The predetermined range from the predetermined position Pl may be a range narrower than the range from the predetermined position Pl to the parking target position P**2**.
[0114] In addition, the controller **11** determines whether or not the obstacle B has been detected in front of the vehicle **1** in the moving direction D based on the external situation acquired by the sensor **14** during the movement by the first autonomous driving.
[0115] For example, the controller **11** analyzes at least one of the sensor information received from the sensor **14** and the captured video captured by the camera **16** by known analysis processing to determine whether or not the obstacle B is present on the teacher route R**1**. Through this determination processing, the controller **11** determines whether or not the obstacle B has been detected ahead in the moving direction D.
[0116] More specifically, the controller **11** determines whether or not the obstacle B is present on the teacher route R**1** in front of the vehicle **1** in the moving direction D by deriving the distance to the obstacle B and the position of the obstacle B by analyzing the detection result of the LiDAR, radar, ultrasonic sensor, or the like included in the sensor **14**, image analysis of the obstacle B included in the captured video, and the like. Through this determination processing, the controller **11** determines whether or not the obstacle B has been detected in front of the vehicle **1** in the moving direction D.
[0117] In the present embodiment, when the controller **11** detects the obstacle B in front of the vehicle **1** in the moving direction D, the driving mode is switched to the temporary manual driving mode, and the operation device **20** and the steering device **27** receive the manual operation by the occupant.
[0118] For example, a scene is assumed in which the controller **11** detects the obstacle B in front of the vehicle **1** in the moving direction D based on the sensor information obtained by the sensor **14** when the vehicle **1** reaches the point S**1** on the teacher route R**1** during the movement of the vehicle **1** by the first autonomous driving along the teacher route R**1** from the predetermined position P**1** to the parking target position P**2**.
[0119] In this case, the controller **11** controls the movement control device **12** to stop the driving of the vehicle **1**, and switches the driving mode from the autonomous driving mode to the temporary manual driving mode. When the controller **11** switches the driving mode from the autonomous driving mode to the temporary manual driving mode, the driving in response to the manual operation by the occupant is temporarily possible during the autonomous driving. That is, in the temporary manual driving mode, the operation device **20** and the steering device **27** receive the manual operation by the occupant, and accordingly, the vehicle **1** is driven in response to the manual operation by the occupant.
[0120] When detecting the obstacle B ahead in the moving direction D, the controller **11** causes the display device **22** to output the third screen. The third screen is a screen prompting the occupant to perform the manual operation for avoiding an obstacle. Specifically, for example, the third screen is a screen including at least one of a phrase prompting the occupant to perform the manual operation for avoiding the obstacle and a schematic diagram prompting the occupant to perform the manual operation for avoiding the obstacle.
[0121] FIGS. **8**A and **8**B are schematic diagrams of an example of a third screen **30**A. The third screen **30**A is an example of a screen **30** output by the display device **22**.
[0122] FIG. **8**A is a schematic diagram of an example of a third screen **30**A**1**. The third screen **30**A**1** is an example of the third screen **30**A.
[0123] The third screen **30**A**1** is, for example, the screen **30** including a message M**1**, a schematic diagram D**1**, and a cancel button **31**. FIG. **8**A illustrates an example in which the message M**1** includes an example of a phrase prompting the occupant to perform a manual operation for avoiding an obstacle: “Stopped due to an obstacle on the registered route. Please move to a position where the obstacle has been avoided through manual driving”. FIG. **8**A illustrates, as an example,

an aspect in which the schematic diagram D**1** is an arrow image representing a recommended driving direction by the manual operation for avoiding an obstacle by the occupant. Note that the message M**1** is not limited to the phrase illustrated in FIG. **8**A as long as the message M**1** includes a phrase prompting the occupant to perform a manual operation to avoid the obstacle. Similarly, the schematic diagram D**1** may be any schematic diagram as long as the schematic diagram prompts the occupant to perform a manual operation for avoiding an obstacle, and is not limited to the aspect illustrated in FIG. **8**A.
[0124] The cancel button **31** is an image area for receiving an instruction to switch from the autonomous driving mode or the temporary manual driving mode to the manual driving mode from the occupant. When the cancel button **31** is operated by the occupant to receive a switching instruction signal to the manual driving mode from the operation device **20** by the operation, the controller **11** switches the driving mode of the vehicle **1** to the manual driving mode. Then, the vehicle **1** controls the movement control device **12** to be driven in response to the manual operation of the occupant. The operation of the cancel button **31** by the occupant is an example of the second operation received by the operation device **20**.
[0125] In addition, the third screen **30**A**1** may further include at least one of the captured video **40**, at least one of a phrase and a schematic diagram representing the current position and posture of the vehicle **1**, at least one of a phrase and a schematic diagram representing the teacher route R**1**, and at least one of a phrase and a schematic diagram representing the obstacle B. The captured video **40** included in the screen **30** may be either a captured video captured by the camera **16** in real time or a captured video captured in the past by the camera **16**.
[0126] FIG. **8**A illustrates, as an example, the third screen **30**A**1** in which an icon representing the vehicle **1** is arranged at a position corresponding to the current position of the vehicle **1** in the captured video **40** to have a posture corresponding to the current posture of the vehicle **1**, and an arrow image representing the teacher route R**1** and an icon representing the obstacle B are superimposed and displayed on the captured video **40**.
[0127] In addition, when the obstacle B is detected ahead in the moving direction D, the controller **11** may further cause a speaker provided in the vehicle **1** to output a voice prompting the occupant to perform the manual operation for avoiding the obstacle.
[0128] FIG. **8**B is a schematic diagram of an example of a third screen **30**A**2**. The third screen **30**A**2** is an example of the third screen **30**A.
[0129] The third screen **30**A**2** further includes an instruction button **32** for receiving an instruction to start manual driving from the occupant on the third screen **30**A**1** illustrated in FIG. **8**A. As described above, the third screen **30**A may further include the instruction button **32**.
[0130] Returning to FIG. **7**, the description will be continued.
[0131] According to the above processing, when the controller **11** detects the obstacle B ahead in the moving direction D during the movement by the first autonomous driving, the third screen **30**A is displayed on the display device **22**. Specifically, when the obstacle B is detected when the vehicle **1** reaches the point S**1** on the teacher route R**1**, the vehicle **1** stops driving at the point S**1**, the autonomous driving mode is switched to the temporary manual driving mode, and the third screen **30**A is output to the display device **22**.
[0132] Then, when receiving the manual operation of the operation device **20** by the occupant, the controller **11** controls the movement control device **12** to be driven in response to the manual operation by the occupant. When the instruction button **32** illustrated in FIG. **8**B is operated by the occupant and a manual operation start signal by the operation is received from the operation device **20**, the controller **11** may control the movement control device **12** to be driven in response to the manual operation by the occupant.
[0133] When the manual operation by the occupant is started from the point S**1** where the obstacle B has been detected, the steering device **27** (steering wheel, handle) or the like is operated by the occupant to avoid the obstacle B, and the vehicle **1** moves by the manual operation by the occupant

(refer to a route R**2** in FIG. **7**).

[0134] While the manual operation is received, the controller **11** causes the display device **22** to output a fourth screen prompting the occupant to perform an operation indicating the avoidance completion when the avoidance of the obstacle B is completed. For example, the fourth screen includes at least one of characters and a schematic diagram prompting the occupant to perform an operation indicating avoidance completion when the avoidance of the obstacle B is completed.

[0135] FIG. **9** is a schematic diagram of an example of a fourth screen **30**B. The fourth screen **30**B is an example of the screen **30** output by the display device **22**.

[0136] The fourth screen **30**B is, for example, the screen **30** including a message M**2** and the cancel button **31**. FIG. **9** illustrates an example in which the message M**2** includes the phrase "Please step on the brake when the avoidance driving is completed" as an example. This phrase is an example of the phrase prompting the occupant to perform an operation indicating the avoidance completion when the avoidance of the obstacle B is completed. In this case, the operation indicating the avoidance completion is "stepping on the brake".

[0137] The fourth screen **30**B may be a screen in which at least the message M**2** and the cancel button **31** are superimposed and displayed on the captured video **40**. In addition, the fourth screen **30**B may further include at least one of the captured video **40**, at least one of a phrase and a schematic diagram representing the current position and posture of the vehicle **1**, at least one of a phrase and a schematic diagram representing the teacher route R**1**, and at least one of a phrase and a schematic diagram representing the obstacle B.

[0138] FIG. **9** illustrates, as an example, the fourth screen **30**B in which an icon representing the vehicle **1** is arranged at a position corresponding to the current position of the vehicle **1** in the captured video **40** to have a posture corresponding to the current posture of the vehicle **1**, and an arrow image representing the teacher route R**1** and an icon representing the obstacle B are superimposed and displayed on the captured video **40**.

[0139] In addition, when the avoidance of the obstacle B has been completed, the controller **11** may further cause a speaker provided in the vehicle **1** to output a voice prompting the occupant to perform an operation indicating the avoidance completion.

[0140] Returning to FIG. **7**, the description will be continued. In the temporary manual driving mode, the controller **11** causes the display device **22** to output the fourth screen **30**B while the operation device **20** receives the manual operation by the occupant. Therefore, the display device **22** switches to the temporary manual driving mode at the point S**1** where the obstacle B has been detected, and outputs the fourth screen **30**B to the display device **22** while the manual operation by the occupant is started and the manual operation is received. That is, in the temporary manual driving mode, the display device **22** outputs the fourth screen **30**B during the movement on the route R**2** through which the vehicle **1** passes while the operation device **20** is operated by the occupant to avoid the obstacle B and the vehicle **1** moves by the manual operation of the occupant. The route R**2** is an example of a movement trajectory of the vehicle **1** moved by the manual operation of the occupant.

[0141] When the obstacle B is detected in front of the vehicle **1** in the moving direction D, the display device **22** outputs the fourth screen **30**B while receiving the manual operation by the occupant, and accordingly, it is possible to prompt the occupant to perform the operation indicating the avoidance completion when the avoidance of the obstacle B is completed by the manual operation.

[0142] Thereafter, in the temporary manual driving mode, the controller **11** causes the display device **22** to output the first screen when the obstacle B is not detected in front of the vehicle **1** in the moving direction D based on the external situation acquired by the sensor **14** after the vehicle **1** is moved by the manual operation by the occupant. The first screen is the screen **30** representing return to the teacher route R**1** by autonomous driving.

[0143] Specifically, when the vehicle **1** starts moving by the manual operation by the occupant, the

controller **11** determines whether or not the obstacle B has been detected in front of the vehicle **1** in the moving direction D based on the external situation acquired by the sensor **14**. The controller **11** may determine whether or not the obstacle B is present in front or the vehicle **1** in the moving direction D in the same manner as described above using the sensor information received from the sensor **14**. When determining that the obstacle B is not detected in front of the vehicle **1** in the moving direction D, the controller **11** causes the display device **22** to output the first screen.
[0144] For example, a scene is assumed in which the operation device **20** is operated by the occupant to avoid the obstacle B when the manual operation by the occupant is started from the point S**1** where the obstacle B is detected, and the vehicle **1** is moved by the manual operation by the occupant (refer to the route R**2** in FIG. **7**). Then, a scene is assumed in which the controller **11** determines that the obstacle B is not detected in front of the vehicle **1** in the moving direction D when the vehicle **1** reaches the point S**2**. In this case, the controller **11** causes the display device **22** to output the first screen.
[0145] The controller **11** may cause the display device **22** to output the first screen when determining that the obstacle B is not detected in front of the vehicle **1** in the moving direction D and receiving a predetermined operation instruction such as stepping on the brake by the occupant.
[0146] FIGS. **10**A and **10**B are schematic diagrams of an example of a first screen **30**C. The first screen **30**C is an example of the screen **30** output by the display device **22**.
[0147] FIG. **10**A is a schematic diagram of an example of a first screen **30**C**1**. The first screen **30**C**1** is an example of the first screen **30**C.
[0148] The first screen **30**C**1** is, for example, the screen **30** including a message M**3** and the cancel button **31**. FIG. **10**A illustrates an example in which the message M**3** includes the phrase “Please release the brake. Return to autonomous driving”. This phrase is an example of a phrase indicating return to the teacher route R**1** by autonomous driving.
[0149] The first screen **30**C**1** may be the screen **30** that includes at least one of a phrase indicating return to the teacher route R**1** by autonomous driving and a schematic diagram indicating return to the teacher route R**1** by autonomous driving.
[0150] In addition, the first screen **30**C**1** may further include at least one of the captured video **40**, at least one of a phrase and a schematic diagram representing the current position and posture of the vehicle **1**, at least one of a phrase and a schematic diagram representing the teacher route R**1**, and at least one of a phrase and a schematic diagram representing the obstacle B.
[0151] FIG. **10**A illustrates, as an example, the first screen **30**C**1** in which an icon representing the vehicle **1** is arranged at a position corresponding to the current position of the vehicle **1** in the captured video **40** to have a posture corresponding to the current posture of the vehicle **1**, and an arrow image representing the teacher route R**1** and an icon representing the obstacle B are superimposed and displayed on the captured video **40**.
[0152] FIG. **10**B is a schematic diagram of an example of a first screen **30**C**2**. The first screen **30**C**2** is an example of the first screen **30**C.
[0153] The first screen **30**C**2** further includes an instruction button **33** for receiving an instruction to start autonomous driving from the occupant on the first screen **30**C**1** illustrated in FIG. **10**A. As described above, the first screen **30**C may further include the instruction button **33**.
[0154] In addition, when the obstacle B is not detected in front of the vehicle **1** in the moving direction D based on the external situation acquired by the sensor **14** after the vehicle **1** is moved by the manual operation, the controller **11** may further cause a speaker provided in the vehicle **1** to output a voice indicating return to the teacher route R**1** by autonomous driving.
[0155] Returning to FIG. **7**, the description will be continued.
[0156] By the above processing, switching to the temporary manual driving mode is performed at the point S**1** where the obstacle B has been detected, the manual operation by the occupant is started, and accordingly, the vehicle **1** moves by the manual operation of the operation device **20** and the steering device **27** by the occupant (refer to the route R**2** in FIG. **7**), and for example, when

it is determined that the obstacle B is not detected ahead in the moving direction D when the vehicle **1** reaches the point S**2**, the display device **22** outputs the first screen **30**C. That is, in the case of the example illustrated in FIG. **7**, for example, when the vehicle **1** which has moved by manual operation reaches the point S**2**, the display device **22** outputs the first screen **30**C.

[0157] By the display device **22** outputting the first screen **30**C, it is possible to provide the occupant in a recognizable manner that the vehicle **1** returns to the teacher route R**1** by autonomous driving when the obstacle B is avoided by the manual operation by the occupant.

[0158] Then, the controller **11** controls the movement control device **12** to perform the second autonomous driving to the halfway point S**3** between the predetermined position P**1** and the parking target position P**2** on the teacher route R**1**, and then perform the third autonomous driving from the halfway point S**3** to the parking target position P**2**.

[0159] The halfway point S**3** may be any point between the predetermined position P**1** and the parking target position P**2** on the teacher route R**1**. Specifically, the halfway point S**3** may be any point between the avoided obstacle B and the parking target position P**2** on the teacher route R**1**.

[0160] The second autonomous driving means autonomous driving from the point S**2** where the obstacle B is avoided by the movement of the vehicle **1** by the manual operation by the occupant and the obstacle B is no longer detected ahead in the moving direction D to the halfway point S**3**. The third autonomous driving means autonomous driving from the halfway point S**3** to the parking target position P**2**.

[0161] The controller **11** may control the movement control device **12** to perform the second autonomous driving based on the difference between the teacher route R**1** and the position of the vehicle **1** at that time. That is, the controller **11** performs feedback control such that the vehicle **1** returns to the teacher route R**1** based on the estimated current position of the vehicle **1** and each position on the teacher route R**1** represented by the teacher route data **18**A. Specifically, the controller **11** controls the movement control device **12** such that the difference between the estimated current position of the vehicle **1** and the position closest to the current position among the positions on the teacher route R**1** represented by the teacher route data **18**A gradually decreases. Furthermore, the controller **11** may perform adjustment to reduce the gain of the feedback control not to cause sudden steering. Furthermore, for example, the controller **11** may perform feedback control not to cause sudden steering by adjusting the correction amount of the difference or the correction amount of the angular deviation to be equal to or less than a predetermined value in a section of a predetermined distance from the start of driving from the point S**2** toward the halfway point S**3**.

[0162] By such control, the controller **11** controls the movement control device **12** such that the vehicle **1** gradually approaches the teacher route R**1** by drawing a trajectory such as the return route R**3**, for example, and reaches the halfway point S**3** by reaching the teacher route R**1**.

[0163] Furthermore, the controller **11** may generate the return route R**3** from the point S**2** to the halfway point S**3** before the start of the second autonomous driving to the halfway point S**3**, and control the movement control device **12** to perform the second autonomous driving based on the return route R**3**.

[0164] FIG. **11** is an explanatory diagram of an example of generation of the return route R**3**. For example, the controller **11** generates a virtual circle C having a radius obtained by adding a predetermined value to the minimum rotation radius of the vehicle **1**, and arranges the circle center of the virtual circle C at a position closest to the current position of the vehicle **1** on the teacher route R**1**. Then, the controller **11** specifies, as the halfway point S**3**, a position on the teacher route R**1** outside the virtual circle C on the teacher route R**1** and in a section E where the route curvature and the change rate are stable. Then, the controller **11** generates, as the return route R**3**, a curve connecting the current position (for example, the point S**2**) of the vehicle **1** to the halfway point S**3** within a predetermined curvature range.

[0165] Note that the controller **11** may specify the halfway point S**3** by the above processing, calculate a route connecting the point S**2** to the halfway point S**3** by a known method, and generate the calculated route as the return route R**3**. In addition, the controller **11** may specify the halfway point S**3** by a known method, calculate a route connecting the point S**2** to the halfway point S**3** by a known method, and generate the calculated route as the return route R**3**.

[0166] Then, when the return route R**3** is generated, the controller **11** may control the movement control device **12** to perform the second autonomous driving along the return route R**3** from the point S**2** toward the halfway point S**3**.

[0167] Returning to FIG. **7**, the description will be continued.

[0168] The controller **11** causes the display device **22** to output the second screen during at least a part of the time while the vehicle **1** is performing the second autonomous driving to the halfway point S**3**. The second screen is the screen **30** representing that the vehicle is in autonomous driving to return to the teacher route R**1**. For example, the second screen includes at least one of a phrase indicating that the vehicle is in autonomous driving to return to the teacher route R**1** and a schematic diagram indicating that the vehicle is in autonomous driving to return to the teacher route R**1**.

[0169] FIGS. **12**A and **12**B are schematic diagrams of an example of a second screen **30**D. The second screen **30**D is an example of the screen **30** output by the display device **22**.

[0170] FIG. **12**A is a schematic diagram of an example of a second screen **30**D**1**. The second screen **30**D**1** is an example of the second screen **30**D.

[0171] The second screen **30**D**1** is, for example, the screen **30** including a message M**4**, a schematic diagram D**2**, and the cancel button **31**. FIG. **12**A illustrates an example in which the message M**4** includes the phrase “Returning to the registered route”. This phrase is an example of a phrase indicating that the vehicle is in autonomous driving to return to the teacher route R**1**. In addition, FIG. **12**A illustrates an aspect in which the schematic diagram D**2** is an arrow image representing that the vehicle is in autonomous driving to return to the teacher route R**1** as an example. Note that the message M**4** only needs to include the phrase indicating that the vehicle is in autonomous driving to return to the teacher route R**1**, and is not limited to the phrase illustrated in FIG. **12**A. Similarly, the schematic diagram D**2** may be any schematic diagram indicating that the vehicle is in autonomous driving to return to the teacher route R**1**, and is not limited to the aspect illustrated in FIG. **12**A.

[0172] In addition, the second screen **30**D**1** may further include at least one of the captured video **40**, at least one of a phrase and a schematic diagram representing the current position and posture of the vehicle **1**, at least one of a phrase and a schematic diagram representing the teacher route R**1**, and at least one of a phrase and a schematic diagram representing the obstacle B. The schematic diagram representing the current position of the vehicle **1** is an example of a second image representing the position of the vehicle **1** at that time. The schematic diagram representing the teacher route R**1** is an example of a first image representing the teacher route R**1**.

[0173] FIG. **12**A illustrates, as an example, the second screen **30**D**1** in which an icon representing the vehicle **1** is arranged at a position corresponding to the current position of the vehicle **1** in the captured video **40** to have a posture corresponding to the current posture of the vehicle **1**, and an arrow image representing the teacher route R**1** and an icon representing the obstacle B are further superimposed and displayed on the captured video **40**.

[0174] When the return route R**3** is generated, the controller **11** may cause the display device **22** to output the second screen **30**D further including the third image representing the generated return route R**3**.

[0175] FIG. **12**B is a schematic diagram of an example of a second screen **30**D**2**. The second screen **30**D**2** is an example of the second screen **30**D.

[0176] As illustrated in FIG. **12**B, when the return route R**3** is generated, the controller **11** generates, for example, a line image representing the return route R**3** as a third image, and causes the display device **22** to output the second screen **30**D**2** in which the third image is further superimposed on the second screen **30**D**1** illustrated in FIG. **12**A. In this case, the display device **22** outputs the second screen **30**D**2** illustrated in FIG. **12**B.

[0177] In addition, the controller **11** may further cause a speaker provided in the vehicle **1** to output a voice indicating that the vehicle **1** is in autonomous driving to return to the teacher route R**1** during at least a part of the time while the vehicle **1** is performing second autonomous driving to the halfway point S**3**.

[0178] Returning to FIG. **7**, the description will be continued.

[0179] The controller **11** controls the movement control device **12** such that the vehicle **1** performs the second autonomous driving from the point S**2** to the halfway point S**3**, and then controls the movement control device **12** to perform the third autonomous driving from the halfway point S**3** to the parking target position P**2** along the teacher route R**1**.

[0180] During at least a part of the third autonomous driving, the controller **11** may cause the display device **22** to output a fifth screen. The fifth screen is the screen **30** representing that the vehicle returns to the teacher route R**1** and continues autonomous driving. Specifically, the fifth screen is the screen **30** including at least one of a phrase and a schematic diagram indicating that the vehicle returns to the teacher route R**1** and continues autonomous driving.

[0181] FIG. **13** is a schematic diagram of an example of a fifth screen **30**E. The fifth screen **30**E is an example of the screen **30**.

[0182] The fifth screen **30**E is, for example, the screen **30** including a message M**5** and the cancel button **31**. FIG. **13** illustrates an example in which the message M**5** includes the phrase “Returned to the registered route. Autonomous driving will continue” as an example. The phrase is an example of a phrase indicating that the vehicle returns to the teacher route R**1** and continues autonomous driving. In addition, the fifth screen **30**E may further include a schematic diagram representing that the vehicle returns to the teacher route R**1** and continues autonomous driving instead of or together with the phrase.

[0183] In addition, the fifth screen **30**E may further include at least one of the captured video **40**, at least one of a phrase and a schematic diagram representing the current position and posture of the vehicle **1**, at least one of a phrase and a schematic diagram representing the teacher route R**1**, and at least one of a phrase and a schematic diagram representing the obstacle B.

[0184] When the vehicle **1** returns to the teacher route R**1** after avoiding the obstacle B by the manual operation by the occupant, the display device **22** outputs the fifth screen **30**E, and accordingly, information indicating that the vehicle **1** returns to the teacher route R**1** and continues autonomous driving can be provided to the occupant.

[0185] Returning to FIG. **7**, the description will be continued.

[0186] On the other hand, in the manual driving mode, when the vehicle **1** is driven in response to the steering operation on the steering device **27** by the occupant and the vehicle **1** is within a predetermined range from the predetermined position P**1**, the operation device **20** may not receive the first operation indicating the instruction to start the autonomous driving mode. In this case, ongoingly in the manual driving mode, the controller **11** drives the vehicle **1** to the parking target position P**2** in response to the steering operation on the steering device **27** by the occupant.

[0187] As described above, the screen **30** (refer to the third screen **30**A to the fifth screen **30**E, and FIGS. **8**A to **13**) displayed in the temporary manual driving mode includes the cancel button **31**. As described above, the cancel button **31** is an image area for receiving an instruction to switch to the manual driving mode from the occupant. In addition, the operation of the cancel button **31** by the occupant is an example of the second operation received by the operation device **20**.

[0188] When the operation device **20** receives the second operation in the temporary manual driving mode, the controller **11** switches the driving mode to the manual driving mode. Then, in the manual driving mode, when the controller **11** does not detect the obstacle B in front of the vehicle **1** in the moving direction based on the external situation acquired by the sensor **14** after the vehicle **1** is moved by manual operation, the controller **11** does not display the first screen **30**C. As described above, the first screen **30**C is the screen **30** representing return to the teacher route R**1** by autonomous driving. Therefore, the first screen **30**C may be displayed in the temporary manual driving mode, but the first screen **30**C is not displayed in the manual driving mode.

[0189] Next, an example of a flow of information processing executed by the controller **11** of the vehicle control device **10** will be described.

[0190] FIG. **14** is a flowchart illustrating an example of a flow of information processing executed by the controller **11** in the teacher driving mode.

[0191] The controller **11** determines whether or not a signal indicating the instruction to start the teacher driving mode has been received from the operation device **20** (step S**100**). When a negative determination is made in step S**100** (step S**100**: No), this routine is ended. When an affirmative determination is made in step S**100** (step S**100**: Yes), the process proceeds to step S**102**.

[0192] When the manual operation of the vehicle **1** by the occupant starts the teacher driving of the vehicle **1**, the controller **11** sequentially stores the map data **18**B and the current position of the vehicle **1** in the storage device **18** (step S**102**). Specifically, the controller **11** sequentially stores the current position of the vehicle **1** sequentially estimated along the teacher driving of the vehicle **1**. At this time, the controller **11** assigns an INDEX to the current position, and sequentially stores the driving position, the azimuth, the driving direction, and the reference driving information, which are the current positions, in association with each other. In addition, the controller **11** specifies feature points by performing image analysis on the captured video captured by the camera **16** during the teacher driving, and sequentially registers the feature points in the map data **18**B.

[0193] The controller **11** determines whether or not the instruction to end the teacher driving mode has been received (step S**104**). The controller **11** determines whether or not a signal indicating the end instruction has been received from the operation device **20** by the operation device **20** by the occupant, thereby making the determination in step S**104**. When a negative determination is made in step S**104** (step S**104**: No), the controller **11** returns to step S**102**. When an affirmative determination is made in step S**104** (step S**104**: Yes), the controller **11** returns to step S**106**.

[0194] In step S**106**, the controller **11** stores the teacher route data **18**A and the map data **18**B in the storage device **18** (step S**106**). The controller **11** sets the driving route R during the teacher driving represented by the group of sequentially estimated current positions stored in the processing of step S**102** as the teacher route R**1**, and stores the teacher route data **18**A representing the teacher route R**1** in the storage device **18**. In addition, the controller **11** stores the map data **18**B in which the feature points specified by the processing in step S**102** are registered in the storage device **18**. Then, this routine is ended.

[0195] FIG. **15** is a flowchart illustrating an example of a flow of information processing executed by the controller **11** in the autonomous driving mode.

[0196] In the manual driving mode, the controller **11** drives the vehicle **1** in response to a steering operation on the steering device **27** by the occupant (step S**200**).

[0197] The controller **11** determines whether or not the position of the vehicle **1** is within a predetermined range from the predetermined position P**1** (step S**202**). When an affirmative determination is made in step S**202** (step S**202**: Yes), the process proceeds to step S**204**.

[0198] The controller **11** determines whether or not the operation device **20** has received the first operation (step S**204**). The controller **11** executes the determination of step S**204** by determining whether or not a signal indicating the instruction to start the autonomous driving mode has been received from the operation device **20**. When an affirmative determination is made in step S**204** (step S**204**: Yes), the process proceeds to step S**206**.

[0199] In step S**206**, the controller **11** switches the driving mode from the manual driving mode to the autonomous driving mode (step S**206**). Then, the controller **11** reads the teacher route data **18**A and the map data **18**B from the storage device **18** (step S**208**). Then, the controller **11** controls the movement control device **12** to start the first autonomous driving from the predetermined position P**1** toward the parking target position P**2** along the teacher route R**1** represented by the teacher route data **18**A (step S**210**).

[0200] When the first autonomous driving is started, the controller **11** determines whether or not the obstacle B has been detected in front of the vehicle **1** in the moving direction D based on the external situation acquired by the sensor **14** (step S**212**). When a negative determination is made in step S**212** (step S**212**: No), the process proceeds to step S**240** to be described later. When an affirmative determination is made in step S**212** (step S**212**: Yes), the process proceeds to step S**214**.

[0201] The controller **11** controls the movement control device **12** to stop driving of the vehicle **1** (step S**214**), and switches the autonomous driving mode to the temporary manual driving mode (step S**216**). When the controller **11** switches the autonomous driving mode to the temporary manual driving mode, driving in response to the manual operation by the occupant is enabled.

[0202] Next, the controller **11** causes the display device **22** to output the third screen **30**A (step S**218**). By the processing in step S**218**, the display device **22** outputs the third screen **30**A illustrated in FIG. **8**A or **8**B, for example.

[0203] The controller **11** determines whether or not the manual operation by the occupant is started (step S**220**). For example, the controller **11** determines whether or not the manual operation of the operation device **20** by the occupant is received, thereby making the determination in step S**220**. When a negative determination is made in step S**220** (step S**220**: No), the controller **11** returns to step S**218**. When it is determined that the manual operation by the occupant is started (step S**220**: Yes), the process proceeds to step S**222**.

[0204] In step S**222**, the controller **11** causes the display device **22** to output the fourth screen **30**B (step S**222**). By the processing in step S**222**, the display device **22** outputs, for example, the fourth screen **30**B illustrated in FIG. **9** to the display device **22**.

[0205] Then, the controller **11** determines whether or not the obstacle B has not been detected in front of the vehicle **1** in the moving direction D based on the external situation acquired by the sensor **14** (step S**224**). When determining that the obstacle B is detected (step S**224**: No), the controller **11** returns to step S**222**. When determining that the obstacle B has not been detected (step S**224**: Yes), the controller **11** proceeds to step S**226**.

[0206] In step S**226**, the controller **11** causes the display device **22** to output the first screen **30**C (step S**226**). When an affirmative determination is made in step S**224** (step S**224**: Yes) and a predetermined operation instruction such as stepping on the brake by the occupant is received, the controller **11** may cause the display device **22** to output the first screen **30**C. By the processing in step S**226**, the display device **22** outputs the first screen **30**C illustrated in FIG. **10**A or **10**B, for example.

[0207] Next, the controller **11** determines whether or not an operation indicating avoidance completion has been received from the occupant (step S**228**), and the controller **11** determines whether or not a predetermined operation such as releasing the leg from the brake by the occupant has been performed, thereby making the determination in step S**228**. When a negative determination is made in step S**228** (step S**282**: No), the controller **11** returns to step S**226**. When an affirmative determination is made in step S**228** (step S**228**: Yes), the controller **11** returns to step S**230**.

[0208] In step S**230**, the controller **11** controls the movement control device **12** to start the second autonomous driving to the halfway point S**3** between the predetermined position P**1** and the parking target position P**2** on the teacher route R**1** (step S**230**).

[0209] When the second autonomous driving is started, the controller **11** causes the display device **22** to output the second screen **30**D (step S**232**). By the processing in step S**232**, the display device **22** outputs the second screen **30**D illustrated in FIG. **12**A or **12**B, for example.

[0210] Then, the controller **11** determines whether or not the vehicle **1** has reached a point on the teacher route R**1**, that is, the halfway point S**3** (step S**234**). For example, the controller **11** determines whether or not the current position of the vehicle **1** is on any position on the teacher route R**1**, thereby making the determination in step S**234**. When a negative determination is made

in step S**234** (step S**234**: No), the controller **11** returns to step S**232**. When an affirmative determination is made in step S**234** (step S**234**: Yes), the controller **11** returns to step S**236**.
[0211] In step S**236**, the controller **11** controls the movement control device **12** to start the third autonomous driving along the teacher route R**1** from the halfway point S**3** to the parking target position P**2** (step S**236**).
[0212] During at least a part of the third autonomous driving, the controller **11** causes the display device **22** to output the fifth screen **30**E (step S**238**). By the processing in step S**234**, for example, the display device **22** outputs the fifth screen **30**E illustrated in FIG. **13**.
[0213] Then, the controller **11** determines whether or not the vehicle **1** has reached the parking target position P**2** (step S**240**). When a negative determination is made in step S**240** (step S**240**: No), the process proceeds to step S**212**. When an affirmative determination is made in step S**240** (step S**240**: Yes), this routine is ended.
[0214] Meanwhile, when a negative determination is made in step S**202** (step S**202**: No), the process proceeds to step S**242**. When a negative determination is made in step S**204** (step S**204**: No), the process proceeds to step S**242**.
[0215] In step S**242**, manual driving for driving the vehicle **1** in response to the steering operation on the steering device **27** by the occupant is continued (step S**242**).
[0216] Next, the controller **11** determines whether or not the obstacle B has been detected in front of the vehicle **1** in the moving direction D based on the external situation acquired by the sensor **14** (step S**244**). When an affirmative determination is made in step S**244** (step S**244**: Yes), the process proceeds to step S**248**. When the obstacle B is not detected (step S**244**: No), the controller **11** does not output the first screen **30**C representing the return to the teacher route R**1** by the autonomous driving to the display device **22** (step S**246**). Then, the process proceeds to step S**248**.
[0217] In step S**248**, the controller **11** determines whether or not the vehicle **1** has reached the parking target position P**2** (step S**248**). When a negative determination is made in step S**248** (step S**248**: No), the process proceeds to step S**202**. When an affirmative determination is made in step S**248** (step S**248**: Yes), this routine is ended.
[0218] Next, an example of a flow of interrupt processing executed between the processing of steps S**216** to S**240** in the information processing illustrated in FIG. **15** by the controller **11** of the present embodiment will be described.
[0219] FIG. **16** illustrates an example of a flow of the interrupt processing executed by the controller **11** of the present embodiment. The controller **11** repeatedly executes the interrupt processing illustrated in FIG. **16** during the processing of steps S**216** to S**240** in the information processing illustrated in FIG. **15**.
[0220] The controller **11** determines whether or not the operation device **20** has received the second operation (step S**300**). The controller **11** determines whether or not the cancel button **31** included in the screen **30** has been operated by the occupant, thereby making the determination in step S**300**. The controller **11** repeats the negative determination (step S**300**: No) until an affirmative determination is made in step S**300** (step S**300**: Yes). When an affirmative determination is made in step S**300** (step S**300**: Yes), the controller **11** switches the driving mode of the vehicle **1** to the manual driving mode (step S**302**). The process returns to step S**242** in FIG. **15**.
[0221] Therefore, when the operation device **20** receives the second operation in the temporary manual driving mode, the driving mode is switched to the manual driving mode.
[0222] The cancel button **31** may be included in the screen **30** displayed when the driving mode is the autonomous driving mode or the temporary manual driving mode. Therefore, in the autonomous driving mode or the temporary manual driving mode, when the occupant operates the cancel button **31** at any timing, the driving mode of the vehicle **1** can be switched from the autonomous driving mode or the temporary manual driving mode to the manual driving mode.
[0223] As described above, the vehicle control device **10** of the present embodiment controls the vehicle **1** including the sensor **14** that acquires the external situation, the operation device **20** that

receives the operation by the occupant, the steering device **27** that receives the steering operation by the occupant, the display device **22** that can be visually recognized by the occupant, and the movement control device **12** that controls at least the steering, the vehicle **1** holding the teacher route R**1** obtained by the teacher driving from the predetermined position P**1** to the parking target position P**2**.

[0224] In the manual driving mode, the vehicle control device **10** drives the vehicle **1** in response to the steering operation on the steering device **27** by the occupant, and when the vehicle **1** is within the predetermined range from the predetermined position P**1**, and the operation device **20** receives the first operation, the vehicle control device **10** switches to the autonomous driving mode. Thereafter, in the autonomous driving mode, the vehicle control device **10** controls at least steering along the teacher route R**1**, and when the obstacle B is detected in front of the vehicle **1** in the moving direction D based on the external situation acquired by the sensor **14** while the vehicle **1** is moving from the predetermined position P**1** toward the parking target position P**2** by the first autonomous driving, the vehicle control device **10** switches to the temporary manual driving mode different from the manual driving mode, and the operation device **20** and the steering device **27** receive the manual operation by the occupant. Thereafter, in the temporary manual driving mode, after the vehicle **1** is moved by the manual operation, when the obstacle B is not detected in front of the vehicle **1** in the moving direction D based on the external situation acquired by the sensor **14**, the display device **22** outputs the first screen **30**C representing the return to the teacher route R**1** by the autonomous driving. The vehicle control device **10** performs the second autonomous driving to the halfway point S**3** between the predetermined position P**1** and the parking target position P**2** on the teacher route R**1**, and then performs the third autonomous driving along the teacher route R**1** from the halfway point S**3** to the parking target position P**2**.

[0225] In the manual driving mode, the vehicle control device **10** drives the vehicle **1** in response to the steering operation on the steering device **27** by the occupant. When the vehicle **1** is within the predetermined range from the predetermined position P**1**, and the operation device **20** does not receive the first operation, in the manual driving mode, the vehicle control device **10** continuously drives the vehicle **1** to the parking target position P**2** in response to the steering operation on the steering device **27** by the occupant.

[0226] In the temporary manual driving mode, the vehicle control device **10** switches to the manual driving mode when the operation device **20** receives the second operation. In the manual driving mode, after the vehicle **1** is moved by the manual operation, when the obstacle B is not detected in front of the vehicle **1** in the moving direction D based on the external situation acquired by the sensor **14**, the display device **22** does not output the first screen **30**C representing the return to the teacher route **41** by the autonomous driving.

[0227] As described above, in the vehicle control device **10** of the present embodiment, in the autonomous driving mode, when the obstacle B is detected in front of the vehicle **1** in the moving direction D while the vehicle **1** is moving from the predetermined position P**1** to the parking target position P**2** by the first autonomous driving along the teacher route R**1**, switching to the temporary manual driving mode different from the manual driving mode is performed, and the operation device **20** receives the manual operation by the occupant. Then, in the vehicle control device **10**, when the obstacle B is not detected in front of the vehicle **1** in the moving direction D after the vehicle **1** moves by the manual operation, the display device **22** outputs the first screen **30**C representing the return to the teacher route R**1** by the autonomous driving. In addition, the vehicle control device **10** performs the second autonomous driving to the halfway point S**3** between the predetermined position P**1** and the parking target position P**2** on the teacher route R**1**, and then performs the third autonomous driving along the teacher route R**1** from the halfway point S**3** to the parking target position P**2**.

[0228] Therefore, in the autonomous driving mode, when the obstacle B that is not present during the teacher driving is present on the teacher route R**1** during the autonomous driving along the

teacher route R**1**, the vehicle control device **10** of the present embodiment can switch to the temporary manual driving mode different from the manual driving mode, prompt the occupant to perform the manual operation for avoiding the obstacle B by the manual operation, and autonomously drive the vehicle **1** from the point S**2** where the obstacle B was avoided toward the parking target position P**2** via the halfway point S**3** on the teacher route R**1**.
[0229] Therefore, the vehicle control device **10** of the present embodiment can provide more suitable parking assistance.
[0230] In addition, according to the vehicle control device **10** of the present embodiment, in the temporary manual driving mode, the occupant performs the manual operation for avoiding the obstacle B, and accordingly, the vehicle **1** after avoiding the obstacle B is driven autonomously from the point (point S**2**) where the obstacle B was avoided toward the parking target position P**2** via the halfway point S**3**. Therefore, the vehicle control device **10** of the present embodiment does not require the driving restart operation or the like by the occupant when the vehicle **1** returns to the teacher route R**1**. Therefore, in addition to the above effect, the vehicle control device **10** of the present embodiment can autonomously drive the vehicle **1** seamlessly from the point S**2** where the obstacle B was avoided toward the parking target position P**2**.
[0231] Next, a hardware configuration of the vehicle control device **10** of the present embodiment will be described.
[0232] FIG. **17** is a block diagram illustrating a hardware configuration example of the vehicle control device **10**.
[0233] The vehicle control device **10** has a hardware configuration using a normal computer in which a central processing unit (CPU) **11**A, a read only memory (ROM) **11**B, a random access memory (RAM) **11**C, an I/F **11**D for connecting to various devices, and the like are connected to each other by a bus **11**E.
[0234] The CPU **11**A is an arithmetic device that controls the entire processing of the vehicle control device **10**. The RAM **11**C stores data necessary for various processing by the CPU **11**A. The ROM **11**B stores programs and the like for realizing various processing by the CPU **11**A. The I/F **11**D is an interface that is connected to an external device or an external terminal via a communication line or the like and transmits and receives data to and from the connected external device or external terminal.
[0235] A program for executing the above-described various processing executed by the vehicle control device **10** is provided by being incorporated in the ROM **11**B or the like in advance. The program for executing the vehicle control method executed in the present embodiment may be provided by being recorded in a computer-readable recording medium such as a CD-ROM, a flexible disk (FD), a CD-R, or a digital versatile disc (DVD) as a file in a format installable or executable in these devices.
[0236] In addition, the program for executing the vehicle control method executed in the present embodiment may be stored on a computer connected to a network such as the Internet and provided by being downloaded via the network. In addition, a program for executing the vehicle control method executed in the present embodiment may be provided or distributed via a network such as the Internet.
[0237] Although embodiments of the present disclosure have been described, the embodiments have been presented by way of example and are not intended to limit the scope of the invention. This embodiment can be implemented in various other aspects, and various omissions, substitutions, and changes can be made without departing from the gist of the invention. As well as being included in the scope and gist of the invention, this embodiment is included in the invention described in the claims and the equivalent scope thereof.

Claims

1. A vehicle control method executed by a vehicle that includes a sensor that acquires an external situation, an operation device that receives an operation by an occupant, a steering device that receives a steering operation by the occupant, a display device that is visually recognizable by the occupant, and a movement controller that controls at least steering, the vehicle holding a teacher route obtained by teacher driving from a predetermined position to a parking target position, wherein in a manual driving mode, the vehicle is driven in response to a steering operation on the steering device by the occupant, and when the vehicle is within a predetermined range from the predetermined position, and the operation device receives a first operation, switching to an autonomous driving mode is performed, thereafter, in the autonomous driving mode, when an obstacle is detected in front of the vehicle in a moving direction based on an external situation acquired by the sensor during movement from the predetermined position to the parking target position along the teacher route by first autonomous driving controlling at least the steering, switching to a temporary manual driving mode different from the manual driving mode is performed, and the operation device and the steering device receive a manual operation by the occupant, thereafter, in the temporary manual driving mode, after the vehicle is moved by the manual operation, when the obstacle is not detected in front of the vehicle in the moving direction based on the external situation acquired by the sensor, the display device outputs a first screen representing return to the teacher route by autonomous driving, after second autonomous driving is performed to a halfway point between the predetermined position and the parking target position on the teacher route, third autonomous driving is performed along the teacher route from the halfway point to the parking target position, in the manual driving mode, the vehicle is driven in response to a steering operation on the steering device by the occupant, and when the vehicle is within a predetermined range from the predetermined position and the operation device does not receive a first operation, ongoingly in the manual driving mode, the vehicle is driven to the parking target position in response to a steering operation on the steering device by the occupant, and in the temporary manual driving mode, when the operation device receives a second operation, switching to the manual driving mode is performed, and in the manual driving mode, after the vehicle is moved by the manual operation, when the obstacle is not detected in front of the vehicle in the moving direction based on an external situation acquired by the sensor, the display device does not output the first screen indicating return to the teacher route by autonomous driving.
2. The vehicle control method according to claim 1, wherein the first screen includes at least one of a phrase indicating return to the teacher route by autonomous driving and a schematic diagram indicating return to the teacher route by autonomous driving.
3. The vehicle control method according to claim 1, wherein the second autonomous driving is performed based on a difference between the teacher route and a position of the vehicle at that time.
4. The vehicle control method according to claim 1, wherein the display device outputs a second screen representing that the vehicle is in autonomous driving to return to the teacher route during at least a part of the time while the vehicle is performing the second autonomous driving to the halfway point.
5. The vehicle control method according to claim 4, wherein the second screen includes at least one of a phrase indicating that the vehicle is in autonomous driving to return to the teacher route and a schematic diagram indicating that the vehicle is in autonomous driving to return to the teacher route.
6. The vehicle control method according to claim 4, wherein the second screen includes a first image representing the teacher route and a second image representing a position of the vehicle at that time.
7. The vehicle control method according to claim 6, wherein in the temporary manual driving mode, after the vehicle is moved by the manual operation, when the obstacle is not detected in front

of the vehicle in the moving direction based on an external situation acquired by the sensor, a return route is generated, and the second autonomous driving is performed based on the return route, and the second screen further includes a third image representing the return route.

8. The vehicle control method according to claim 1, wherein when an obstacle is detected in front of the vehicle in the moving direction based on an external situation acquired by the sensor, the display device displays a third screen prompting the occupant to perform a manual operation for avoiding the obstacle.

9. The vehicle control method according to claim 8, wherein the third screen includes at least one of a phrase prompting the occupant to perform a manual operation for avoiding an obstacle, and a schematic diagram for prompting the occupant to perform a manual operation for avoiding an obstacle.

10. The vehicle control method according to claim 1, wherein in the autonomous driving mode, when an obstacle is detected in front of the vehicle in the moving direction based on an external situation acquired by the sensor during movement from the predetermined position toward the parking target position along the teacher route by first autonomous driving controlling at least the steering, switching to a temporary manual driving mode different from the manual driving mode is performed, and while the operation device receives the manual operation by the occupant, the display device outputs a fourth screen prompting the occupant to perform an operation indicating avoidance completion when avoidance of the obstacle is completed.

11. A vehicle control device that controls a vehicle that includes a sensor that acquires an external situation, an operation device that receives an operation by an occupant, a steering device that receives a steering operation by the occupant, a display device that is visually recognizable by the occupant, and a movement controller that controls at least steering, the vehicle holding a teacher route obtained by teacher driving from a predetermined position to a parking target position, wherein in a manual driving mode, the vehicle is driven in response to a steering operation on the steering device by the occupant, and when the vehicle is within a predetermined range from the predetermined position, and the operation device receives a first operation, switching to an autonomous driving mode is performed, thereafter, in the autonomous driving mode, when an obstacle is detected in front of the vehicle in a moving direction based on an external situation acquired by the sensor during movement from the predetermined position to the parking target position along the teacher route by first autonomous driving controlling at least the steering, switching to a temporary manual driving mode different from the manual driving mode is performed, and the operation device and the steering device receive a manual operation by the occupant, thereafter, in the temporary manual driving mode, after the vehicle is moved by the manual operation, when the obstacle is not detected in front of the vehicle in the moving direction based on the external situation acquired by the sensor, the display device outputs a first screen representing return to the teacher route by autonomous driving, after second autonomous driving is performed to a halfway point between the predetermined position and the parking target position on the teacher route, third autonomous driving is performed along the teacher route from the halfway point to the parking target position, in the manual driving mode, the vehicle is driven in response to a steering operation on the steering device by the occupant, and when the vehicle is within a predetermined range from the predetermined position and the operation device does not receive a first operation, ongoingly in the manual driving mode, the vehicle is driven to the parking target position in response to a steering operation on the steering device by the occupant, and in the temporary manual driving mode, when the operation device receives a second operation, switching to the manual driving mode is performed, and in the manual driving mode, after the vehicle is moved by the manual operation, when the obstacle is not detected in front of the vehicle in the moving direction based on an external situation acquired by the sensor, the display device does not output the first screen indicating return to the teacher route by autonomous driving.

12. The vehicle control device according to claim 11, wherein the first screen includes at least one

of a phrase indicating return to the teacher route by autonomous driving and a schematic diagram indicating return to the teacher route by autonomous driving.
13. The vehicle control device according to claim 11, wherein the second autonomous driving is performed based on a difference between the teacher route and a position of the vehicle at that time.
14. The vehicle control device according to claim 11, wherein the display device outputs a second screen representing that the vehicle is in autonomous driving to return to the teacher route during at least a part of the time while the vehicle is performing the second autonomous driving to the halfway point.
15. The vehicle control device according to claim 14, wherein the second screen includes at least one of a phrase indicating that the vehicle is in autonomous driving to return to the teacher route and a schematic diagram indicating that the vehicle is in autonomous driving to return to the teacher route.
16. The vehicle control device according to claim 14, wherein the second screen includes a first image representing the teacher route and a second image representing a position of the vehicle at that time.
17. The vehicle control device according to claim 16, wherein in the temporary manual driving mode, after the vehicle is moved by the manual operation, when the obstacle is not detected in front of the vehicle in the moving direction based on an external situation acquired by the sensor, a return route is generated, and the second autonomous driving is performed based on the return route, and the second screen further includes a third image representing the return route.
18. The vehicle control device according to claim 11, wherein when an obstacle is detected in front of the vehicle in the moving direction based on an external situation acquired by the sensor, the display device displays a third screen prompting the occupant to perform a manual operation for avoiding the obstacle.
19. The vehicle control device according to claim 18, wherein the third screen includes at least one of a phrase prompting the occupant to perform a manual operation for avoiding an obstacle, and a schematic diagram for prompting the occupant to perform a manual operation for avoiding an obstacle.
20. The vehicle control device according to claim 11, wherein in the autonomous driving mode, when an obstacle is detected in front of the vehicle in the moving direction based on an external situation acquired by the sensor during movement from the predetermined position toward the parking target position along the teacher route by first autonomous driving controlling at least the steering, switching to a temporary manual driving mode different from the manual driving mode is performed, and while the operation device receives the manual operation by the occupant, the display device outputs a fourth screen prompting the occupant to perform an operation indicating avoidance completion when avoidance of the obstacle is completed.